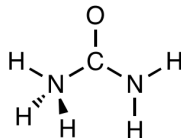
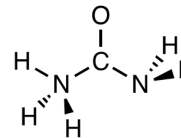
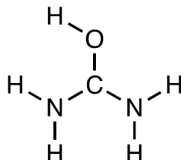
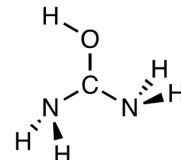
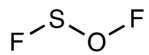
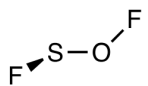
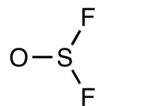
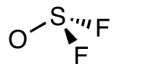
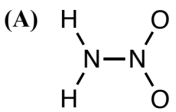
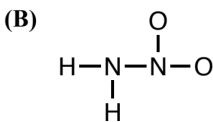
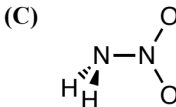
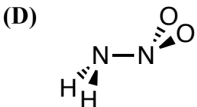
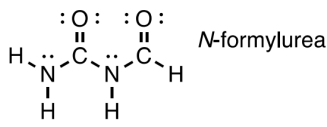
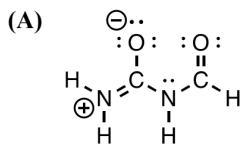
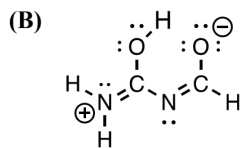
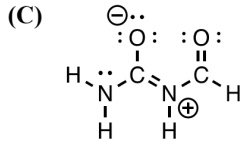
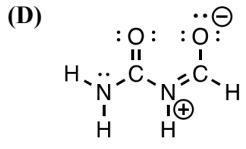


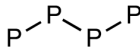
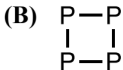
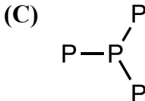
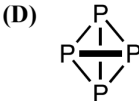
Bonding/Molecular Structure

- In which are the species arranged in increasing order of N-O bond length?
 (A) $\text{NO}^+ < \text{NO} < \text{NO}_3^-$ (B) $\text{NO}^+ < \text{NO}_3^- < \text{NO}$
 (C) $\text{NO} < \text{NO}^+ < \text{NO}_3^-$ (D) $\text{NO}_3^- < \text{NO}^+ < \text{NO}$
- Which statement does not accurately describe σ and π bonding?
 (A) σ bonds have no nodes containing the internuclear axis, while π bonds have one node containing the internuclear axis.
 (B) σ and π bonds are of similar strength in homodiatom molecules containing elements of the second and third period.
 (C) Triple bonds in heterodiatom molecules of the second and third period elements consist of one σ and two π bonds.
 (D) Formaldehyde, H_2CO , has three σ bonds and one π bond.
- What is the bond order in superoxide ion, O_2^- ?
 (A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
- Which is the best description of the geometry of XeOF_4 ?
 (A) Square pyramidal with the oxygen axial
 (B) Square pyramidal with the oxygen equatorial
 (C) Trigonal bipyramidal with the oxygen axial
 (D) Trigonal bipyramidal with the oxygen equatorial
- Which is the best description of the arrangement of the atoms in space in the protonated urea ion, $\text{H}_5\text{CN}_2\text{O}^+$?
 (A)  (B) 
 (C)  (D) 
- How many stereoisomers are there of the square planar complex $(\text{en})\text{PtClBr}$ ($\text{en} = 1,2\text{-ethylenediamine}$, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$)?
 (A) 1 (B) 2 (C) 3 (D) 4
- Which molecule is trigonal planar?
 (A) NH_3 (B) NF_3 (C) SO_3 (D) ClF_3
- Which molecule has the shortest N-O bond?
 (A) NO (B) NO_2 (C) N_2O (D) N_2O_4
- How many lone pairs are there in the Lewis structure of nitromethane, CH_3NO_2 ?
 (A) 3 (B) 4 (C) 5 (D) 7
- Which is the strongest acid in aqueous solution?
 (A) NH_4^+ (B) H_2O (C) PH_4^+ (D) H_2S
- Which is the best representation of the three-dimensional arrangement of the atoms in thionyl fluoride, SOF_2 ?
 (A)  (B) 
 (C)  (D) 
- Which ion is diamagnetic?
 (A) $\text{Fe}(\text{CN})_6^{4-}$ (B) CoCl_4^{2-}
 (C) $\text{Ni}(\text{H}_2\text{O})_6^{2+}$ (D) $\text{Cu}(\text{NH}_3)_4^{2+}$
- Which species has the shortest bond to nitrogen?
 (A) HCN (B) N_2O_4 (C) NO_2 (D) NO_2^-
- How many lone pairs are there in a valid Lewis structure of ozone, O_3 ?
 (A) 6 (B) 7 (C) 8 (D) 9
- Which species is linear?
 (A) H_2F^+ (B) FCO^+ (C) FNO (D) N_2F_2
- Anionic NO^- has a longer bond than neutral NO . Which is the best explanation for this difference?
 (A) NO^- has two unpaired electrons while NO has only one.
 (B) NO^- has a double bond while NO has a triple bond.
 (C) NO^- has one more σ antibonding electron than NO .
 (D) NO^- has one more π antibonding electron than NO .

Bonding/Molecular Structure

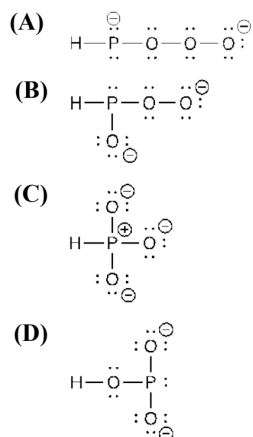
17. Which is the best representation of the three-dimensional arrangement of the atoms in nitramide, H_2NNO_2 ?
- (A)  (B) 
- (C)  (D) 
18. How many stereoisomers are possible for the square planar complex $\text{Pt}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)(\text{C}_2\text{O}_4)$?
- (A) 1 (B) 2 (C) 3 (D) 4
19. Which species has the largest O–N–O angle?
- (A) FNO_2 (B) NO_2 (C) NO_2^- (D) NO_3^-
20. Which molecule has the shortest bond?
- (A) H_2 (B) N_2 (C) O_2 (D) F_2
21. Which molecules are described by molecular orbital theory as having a bond order of 2.0?
- I. CO II. NO
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
22. Which is the best description of the geometry of IF_7 ?
- (A) Heptagonal planar
(B) Hexagonal monopyrarnidal
(C) Pentagonal bipyramidal
(D) Capped trigonal prismatic
23. Carbon monoxide, CO, has a very small dipole moment. Which is the best explanation for this observation?
- (A) The carbon-oxygen π bonds are polarized in the opposite direction from the carbon-oxygen σ bonds, so the bond dipoles tend to cancel each other out.
(B) Carbon and oxygen each have the same number of lone pairs despite the fact that carbon has fewer valence electrons than oxygen, partially canceling the bond dipoles.
(C) Neither carbon nor oxygen has a formal charge in the Lewis structure of CO, giving an electron distribution that is minimally asymmetric.
(D) Carbon is sp -hybridized in CO, making it more electronegative than a typical carbon atom.
24. Which species has the smallest tendency to dimerize?
- (A) NO (B) NO_2 (C) CO_2^- (D) SO_2^-
25. Which molecule has the strongest bond?
- (A) H_2 (B) N_2 (C) O_2 (D) F_2
26. What is the geometry of the iodate ion, IO_3^- ?
- (A) Trigonal pyramidal (B) Trigonal planar
(C) T-shaped (D) Zigzag
27. Which is NOT a valid resonance structure of *N*-formylurea?
-  *N*-formylurea
- (A)  (B) 
- (C)  (D) 
28. Which molecule is best described as having a lone pair on nitrogen in an sp^2 -hybridized orbital?
- (A) NH_3 (B) HCN
(C) H_2CNH (D) H_3CNH_2
29. Nitric oxide, NO, has a smaller first ionization energy than either N atoms or O atoms. Which is the best explanation for this observation?
- (A) The electron that is ionized in NO occupies an antibonding orbital.
(B) NO forms a strong bond, which lowers the ionization energy.
(C) NO has a large dipole moment, which increases electron-electron repulsion.
(D) N atoms have a half-filled $2p$ subshell, while O is very electronegative. Both of these factors give the isolated atoms unusually high ionization energies.
30. How many stereoisomers does the octahedral complex $\text{Co}(\text{NH}_3)_3(\text{CN})_3$ have?
- (A) 1 (B) 2 (C) 3 (D) 4

Bonding/Molecular Structure

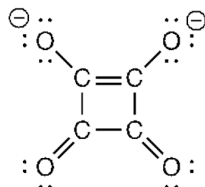
31. Which molecule has a nonzero dipole moment?
(A) CF_4 (B) C_2F_4 (C) SF_4 (D) SF_6
32. How many π bonds are in cyanogen, NCCN ?
(A) 2 (B) 4 (C) 6 (D) 7
33. In azomethane, H_3CNNCH_3 , what are the molecular geometries around the carbon and nitrogen atoms, respectively?
(A) Tetrahedral at carbon, bent at nitrogen
(B) Tetrahedral at carbon, linear at nitrogen
(C) Square planar at carbon, bent at nitrogen
(D) Square planar at carbon, linear at nitrogen
34. What is the bond order of O_2^{+} ?
(A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
35. Which is the best explanation for phosphorus favoring formation of P_4 over P_2 molecules, while nitrogen forms only N_2 molecules?
(A) Phosphorus-phosphorus π bonds are weaker than nitrogen-nitrogen π bonds.
(B) Phosphorus can have an expanded octet while nitrogen cannot.
(C) The nonbonding electrons on phosphorus occupy an unhybridized s orbital while the nonbonding electrons on nitrogen occupy hybrid orbitals.
(D) The larger size of phosphorus allows the bonding electrons to be closer together.
36. In the octahedral complex $\text{Co}(\text{NH}_2\text{CH}_2\text{CO}_2)_3$, each glycinate ion $\text{NH}_2\text{CH}_2\text{CO}_2^-$ binds to cobalt through its nitrogen atom and one of its oxygen atoms. How many stereoisomers of $\text{Co}(\text{NH}_2\text{CH}_2\text{CO}_2)_3$ are possible?
(A) 2 (B) 4 (C) 6 (D) 8
37. Which species has the strongest carbon-oxygen bond?
(A) CO (B) CO_2
(C) CH_2O (D) CH_3OH
38. Which molecule has a nonzero dipole moment?
(A) O_2 (B) O_3 (C) S_8 (D) SO_3
39. Which statement best describes the structure and bonding in nitromethane, $\text{H}_3\text{C}-\text{NO}_2$?
(A) Each of the three bonds to nitrogen is a different length.
(B) The molecule is Lewis acidic because the nitrogen does not obey the octet rule.
(C) The molecule is Lewis basic because the nitrogen has a lone pair.
(D) The nitrogen atom has a trigonal planar geometry.
40. Which statements about the superoxide ion, O_2^- , are correct?
I. It has a bond order of 2.0.
II. It has exactly one unpaired electron.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
41. Which structure best depicts the three-dimensional shape of the P_4 molecule?
(A)  (B) 
(C)  (D) 
42. How many isomers are there of the octahedral coordination complex $\text{Co}(\text{NH}_3)_3\text{Cl}_3$?
(A) 1 (B) 2 (C) 3 (D) 4
43. Which molecule is planar?
(A) CF_4 (B) COF_2 (C) SF_4 (D) SOF_2
44. Which molecule has no unpaired electrons?
(A) NO (B) O_2 (C) SO_2 (D) ClO_2

Bonding/Molecular Structure

45. Phosphorous acid, H_3PO_3 , is a diprotic acid in aqueous solution ($\text{p}K_1 = 2.0$, $\text{p}K_2 = 6.6$). Which Lewis structure best represents HPO_3^{2-} ?



46. One Lewis structure of squarate ion, $\text{C}_4\text{O}_4^{2-}$, is shown below. Which statement best describes the bond distances in the squarate ion?



- (A) All carbon-carbon bond distances are the same, and all carbon-oxygen bond distances are the same.
- (B) There are two distinct carbon-carbon bond distances, but all carbon-oxygen bond distances are the same.
- (C) There are two distinct carbon-carbon bond distances, and two distinct carbon-oxygen bond distances.
- (D) There are three distinct carbon-carbon bond distances, and two distinct carbon-oxygen bond distances.
47. The bond distance in CO is 112.8 pm, while the bond distance in NO is 115 pm. Which is the best explanation for the shorter bond in CO?
- (A) Carbon has one fewer valence electron than nitrogen.
- (B) Carbon is less electronegative than nitrogen.
- (C) Carbon has a smaller atomic radius than nitrogen.
- (D) Carbon can form up to four bonds, while nitrogen can form only three.
48. How many stereoisomers does the square planar complex $\text{Pt}(\text{NH}_3)_2(\text{Br})(\text{Cl})$ have?
- (A) One (B) Two (C) Three (D) Four

49. Which statements correctly describe the geometry of the carbonate ion, CO_3^{2-} ?

I. All three carbon-oxygen bond distances are the same.
II. All three bond angles are 120° .

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

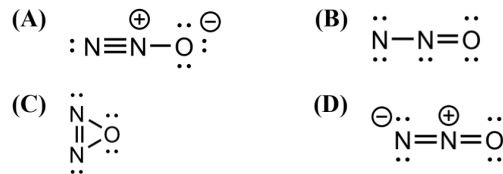
50. Which molecule has a trigonal pyramidal geometry?

- (A) PCl_3 (B) BCl_3 (C) IF_3 (D) SO_3

51. The bond in F_2 has a smaller bond dissociation enthalpy than the bond in Cl_2 . Which is the best explanation for this difference?

- (A) Fluorine is more electronegative than chlorine.
(B) Lone pair-lone pair repulsions are stronger in F_2 than in Cl_2 .
(C) The F-F bond is longer than the Cl-Cl bond.
(D) Chlorine has energetically accessible d orbitals while fluorine does not.

52. Which resonance structure contributes the most to the overall bonding in nitrous oxide, N_2O ?



53. How many σ bonds and how many π bonds are present in allene, $\text{H}_2\text{C}=\text{C}=\text{CH}_2$?

- (A) One σ , one π (B) Five σ , one π
(C) Six σ , two π (D) Seven σ , two π

54. Which aspect of metallic bonding best accounts for the characteristic malleability of metals?

- (A) Metals have few valence electrons.
(B) Metals contain a lattice of positive ions.
(C) Metals contain delocalized electrons.
(D) Metals have low electronegativity.

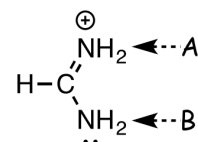
55. Which compound contains both ionic and covalent bonds?

- (A) PF_3 (B) KF
(C) CH_3COOH (D) MgSO_4

Bonding/Molecular Structure

56. Which gas-phase molecule is NOT linear?
 (A) CS₂ (B) SO₂ (C) HCCH (D) BrCN
57. In the Lewis structure of the chlorate ion, ClO₃⁻, how many lone pairs of electrons does the chlorine atom have?
 (A) 0 (B) 1 (C) 2 (D) 3
58. A coordination complex M(NH₃)₂Cl₂ can be separated into a pair of geometric isomers. Is this observation consistent with a tetrahedral or a square planar geometry at the metal center?
 (A) It is consistent only with a tetrahedral geometry.
 (B) It is consistent only with a square planar geometry.
 (C) It is consistent with either a square planar or a tetrahedral geometry.
 (D) It is consistent with neither a square planar nor a tetrahedral geometry.
59. The bond in gas-phase O₂ (121 pm) is significantly longer than the bond in gas-phase O₂⁺ (112 pm). What is the best explanation for this difference?
 (A) O₂ has one more antibonding electron than O₂⁺.
 (B) O₂ has two unpaired electrons while O₂⁺ has one.
 (C) The bond in O₂ has less ionic character than the bond in O₂⁺.
 (D) It requires more energy to remove an electron from O₂ to form O₂⁺ than it does to remove an electron from O to form O⁺.
60. In the guanidinium ion, [C(NH₂)₃]⁺, what is the best description of the hybridizations of the nitrogen atoms?
 (A) All three sp³ (B) Two sp³, one sp²
 (C) One sp³, two sp² (D) All three sp²
61. Which molecule has a dipole moment of zero?
 (A) CO (B) CO₂ (C) CH₂O (D) CH₃OH
62. What is the geometry of SF₄?
 (A) Tetrahedral
 (B) Square planar
 (C) See-saw
 (D) Trigonal monopyramidal

63. How many σ and π bonds are in 1,3-butadiene, H₂C=CH-CH=CH₂?
 (A) 7 σ and 2 π bonds (B) 2 σ and 7 π bonds
 (C) 9 σ and 2 π bonds (D) 2 σ and 9 π bonds
64. What is the formal charge on the central nitrogen in the Lewis structure of N₂O?
 (A) 0
 (B) +1
 (C) -1
 (D) 0 in some resonance structures, -1 in other resonance structures
65. Shown below is a Lewis structure of the formamidinium ion. What geometry is exhibited by the two nitrogens in the formamidinium ion?



<u>Geometry at A</u>	<u>Geometry at B</u>
(A) trigonal planar	trigonal planar
(B) trigonal planar	trigonal pyramidal
(C) trigonal pyramidal	trigonal planar
(D) trigonal pyramidal	trigonal pyramidal

66. How many unpaired electrons are in the superoxide ion, O₂⁻?
 (A) Zero (B) One (C) Two (D) Three
67. Which molecule is nonpolar, yet contains polar covalent bonds?
 (A) CO₂ (B) HCN (C) NH₃ (D) P₄
68. The nitrite ion, NO₂⁻, can be represented as a resonance hybrid of two significant Lewis structures. Which statements about this are correct?
 I. The two resonance structures contribute equally to the structure.
 II. The formal charge of nitrogen is zero in both resonance structures.
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

Bonding/Molecular Structure

69. Which is not a stable molecule?
(A) NF_3 (B) NF_5 (C) PF_3 (D) PF_5
70. What is the molecular geometry of IF_3 ?
(A) Trigonal planar (B) Trigonal pyramidal
(C) T-shaped (D) Tetrahedral
71. N_2 is a stable molecule and the N_4 molecule is unknown. P_4 is much more stable than molecular P_2 . Which is the best explanation for this difference?
(A) N_2 has valence electrons only in bonding and nonbonding orbitals, while P_2 has some valence electrons in antibonding orbitals.
(B) The greater electronegativity of N compared to P stabilizes compounds with lower molar masses.
(C) The greater size of P compared to N results in decreased overlap in pi bonds.
(D) The preference of P to adopt smaller bond angles than N favors formation of tetrahedral P_4 molecules.
72. Which contains sp^3 -hybridized carbon atoms?
(A) Benzene, C_6H_6 (B) Ethane, C_2H_6
(C) Ethene, C_2H_4 (D) Ethyne, C_2H_2